



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

Class: FYMCA Semester: I	Div: B	Course Code: MCA01504 Course Name: Python Programming Laboratory	Batch: F2
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CO No: 3		Assignment No: 7	

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Title: Create an application for department library which has N books write functions for the following:

- Delete the duplicate entries
- Display books in ascending order based on cost of books
- Count the number of books with a cost of 500.

Code:

```
def deleteDuplicates(library):
    seen = set()
    unique_library = list()
    for book in library:
        bookName = book["title"]
        if bookName not in seen:
            seen.add(bookName)
            unique_library.append(book)

    return unique_library

def sortByCost(library):
    return sorted(library, key=lambda l: l['cost'])

def count500(library):
    cnt = 0;
    for book in library:
        if book['cost'] == 500:
            cnt += 1;
    return cnt

def printLibrary(msg, library):
    print(msg)
    for book in library:
        print(f"name = {book['title']} ---- cost = {book['cost']}")

library = [
    {"title": "Programming in Python", "author": "Dr. Rizwan rehman", "cost": 500},
    {"title": "Data Structures and Algorithms", "author": "Prof. Naveen Garg", "cost": 300},
    {"title": "Programming in Python", "author": "Dr. Rizwan rehman", "cost": 500},
    {"title": "Internet of Things: A Hands-On Approach", "author": "Vijay Wadiseti", "cost": 700},
    {"title": "Research Methodology", "author": "Ranjit Kumar", "cost": 500},
```



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```
["title": "Elements of Discrete Mathematics", "author": "C. L. Liu", "cost": 300},  
{"title": "Internet of Things: A Hands-On Approach", "author": "Vijay Wadiseti", "cost": 700},  
{"title": "Internet of Things: A Hands-On Approach", "author": "Vijay Wadiseti", "cost": 700},  
]
```

```
printLibrary("Current Library:", library)
```

```
library = deleteDuplicates(library)  
printLibrary("\nDeleting Duplicates\nUpdated Library:", library)  
printLibrary("\nSorting by Cost\nSort library:", sortByCost(library))  
print("\nCounting books\n...", count500(library), "books has cost of 500")
```

Output:

```
Current Library:  
name = Programming in Python ---- cost = 500  
name = Data Structures and Algorithms ---- cost = 300  
name = Programming in Python ---- cost = 500  
name = Internet of Things: A Hands-On Approach ---- cost = 700  
name = Research Methodology ---- cost = 500  
name = Elements of Discrete Mathematics ---- cost = 300  
name = Internet of Things: A Hands-On Approach ---- cost = 700  
name = Internet of Things: A Hands-On Approach ---- cost = 700  
  
Deleting Duplicates  
Updated Library:  
name = Programming in Python ---- cost = 500  
name = Data Structures and Algorithms ---- cost = 300  
name = Internet of Things: A Hands-On Approach ---- cost = 700  
name = Research Methodology ---- cost = 500  
name = Elements of Discrete Mathematics ---- cost = 300  
  
Sorting by Cost  
Sort library:  
name = Data Structures and Algorithms ---- cost = 300  
name = Elements of Discrete Mathematics ---- cost = 300  
name = Programming in Python ---- cost = 500  
name = Research Methodology ---- cost = 500  
name = Internet of Things: A Hands-On Approach ---- cost = 700  
  
Counting books  
... 2 books has cost of 500
```